

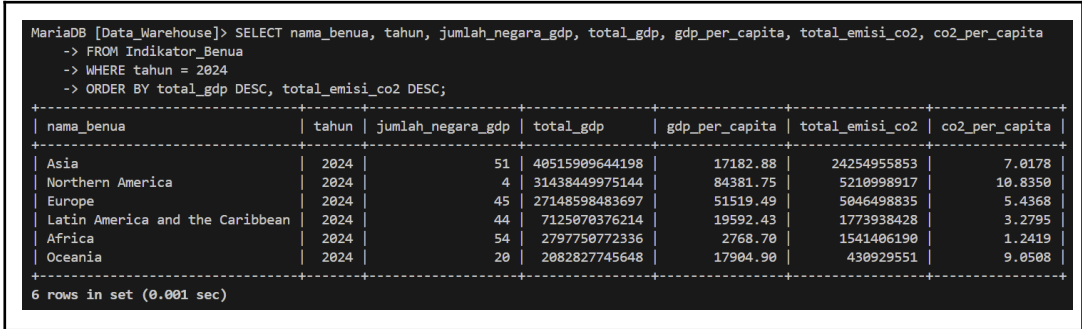
QUERY Data Warehouse, Optimization Query, & ss scheduling

1. Perbandingan ekonomi dan emisi tiap benua pada tahun 2024

QUERY:

```
SELECT nama_benua, tahun, jumlah_negara_gdp, total_gdp,
gdp_per_capita, total_emisi_co2, co2_per_capita
FROM Indikator_Benua
WHERE tahun = 2024
ORDER BY total_gdp DESC, total_emisi_co2 DESC;
```

SCREENSHOT:



```
MariaDB [Data_Warehouse]> SELECT nama_benua, tahun, jumlah_negara_gdp, total_gdp, gdp_per_capita, total_emisi_co2, co2_per_capita
-> FROM Indikator_Benua
-> WHERE tahun = 2024
-> ORDER BY total_gdp DESC, total_emisi_co2 DESC;
```

nama_benua	tahun	jumlah_negara_gdp	total_gdp	gdp_per_capita	total_emisi_co2	co2_per_capita
Asia	2024	51	40515909644198	17182.88	24254955853	7.0178
Northern America	2024	4	31438449975144	84381.75	5210998917	10.8350
Europe	2024	45	27148598483697	51519.49	5046498835	5.4368
Latin America and the Caribbean	2024	44	7125070376214	19592.43	1773938428	3.2795
Africa	2024	54	2797750772336	2768.70	1541406190	1.2419
Oceania	2024	20	2082827745648	17904.90	430929551	9.0508

6 rows in set (0.001 sec)

2. Hubungan kategori ekonomi dengan tingkat urbanisasi & emisi

QUERY:

```
SELECT jenis_kategori_ekonomi, COUNT(id_negara) AS jumlah_negara,
AVG(persentase_penduduk_urban) AS avg_persentase_urban,
AVG(usia_median) AS avg_usia_median,
AVG(emisi_co2_per_capita) AS avg_emisi_co2_per_capita
FROM Indikator_Negara
WHERE tahun = 2024 AND jenis_kategori_ekonomi IS NOT NULL
GROUP BY jenis_kategori_ekonomi
ORDER BY avg_emisi_co2_per_capita DESC, avg_persentase_urban DESC;
```

SCREENSHOT:

MariaDB [Data_Warehouse]> SELECT jenis_kategori_ekonomi,COUNT(id_negara) AS jumlah_negara,
-> AVG(persentase_penduduk_urban) AS avg_persentase_urban,
-> AVG(usia_median) AS avg_usia_median,
-> AVG(emisi_co2_per_capita) AS avg_emisi_co2_per_capita
-> FROM Indikator_Negara
-> WHERE tahun = 2024 AND jenis_kategori_ekonomi IS NOT NULL
-> GROUP BY jenis_kategori_ekonomi
-> ORDER BY avg_emisi_co2_per_capita DESC, avg_persentase_urban DESC;

jenis_kategori_ekonomi	jumlah_negara	avg_persentase_urban	avg_usia_median	avg_emisi_co2_per_capita
High income	89	73.357143	39.535955	8.631688
Upper middle income	54	65.450943	30.938889	3.856327
Lower middle income	48	46.246809	22.833333	1.149348
Low Income	27	39.255556	18.511111	0.346923

4 rows in set (0.026 sec)

3. Top 10 negara dengan perbandingan konsumsi energi non fosil / fosil tertinggi
(lebih banyak menggunakan energi terbarukan)

QUERY:
SELECT n.nama_negara, k.tahun,
SUM(CASE WHEN j.kategori = 'Fosil' THEN k.jumlah_komposisi ELSE
0 END) AS konsumsi_energi_fosil,
SUM(CASE WHEN j.kategori = 'Non-Fosil' THEN k.jumlah_komposisi
ELSE 0 END) AS konsumsi_energi_non_fosil,
ROUND(SUM(CASE WHEN j.kategori = 'Non-Fosil' THEN
k.jumlah_komposisi ELSE 0 END) * 100 /
SUM(k.jumlah_komposisi), 2) AS persentase_perbandingan_energi
FROM Komposisi_Energi k
JOIN Negara n ON k.id_negara = n.id_negara
JOIN Jenis_Energi j ON k.id_jenis = j.id_jenis
GROUP BY n.nama_negara, k.tahun
ORDER BY persentase_perbandingan_energi DESC
LIMIT 10;

SCREENSHOT:

```

MariaDB [Data_Warehouse]> SELECT n.nama_negara, k.tahun,
-> SUM(CASE WHEN j.kategori = 'Fosil' THEN k.jumlah_komposisi ELSE
-> 0 END) AS konsumsi_energi_fosil,
-> SUM(CASE WHEN j.kategori = 'Non-Fosil' THEN k.jumlah_komposisi
-> ELSE 0 END) AS konsumsi_energi_non_fosil,
-> ROUND(SUM(CASE WHEN j.kategori = 'Non-Fosil' THEN
-> k.jumlah_komposisi ELSE 0 END) * 100 /
-> SUM(k.jumlah_komposisi), 2) AS persentase_perbandingan_energi
-> FROM Komposisi_Energi k JOIN Negara n ON k.id_negara = n.id_negara JOIN Jenis_Energi j ON k.id_jenis = j.id_jenis
-> GROUP BY n.nama_negara, k.tahun
-> ORDER BY persentase_perbandingan_energi DESC
-> LIMIT 10;

```

nama_negara	tahun	konsumsi_energi_fosil	konsumsi_energi_non_fosil	persentase_perbandingan_energi
Falkland Islands	2017	0	82900000	100.00
Réunion	2017	0	9771585000	100.00
Martinique	2017	0	707538000	100.00
Guadeloupe	2017	0	2010517000	100.00
Kiribati	2017	0	46100000	100.00
French Guiana	2017	0	5699060000	100.00
Bhutan	2017	6685155000	54597299000	89.09
Iceland	2017	46699237000	174750255000	78.91
Paraguay	2017	121460000000	397226518000	76.58
Norway	2017	610235682000	1291138873000	67.91

10 rows in set (0.005 sec)

4. Mengambil data per dekade (*hanya ada 2020s*)

QUERY:

```

SELECT t.dekade, COUNT(n.id_negara) AS
  Jumlah_negara, SUM(n.nilai_gdp) AS total_gdp,
  SUM(n.emisi_co2) AS total_emisi_co2
FROM Indikator_Negara n
JOIN Tahun t ON n.tahun = t.tahun
GROUP BY t.dekade
ORDER BY t.dekade ASC;

```

SCREENSHOT:

Before Insert

```

MariaDB [Data_Warehouse]> SELECT t.dekade, COUNT(n.id_negara) AS
-> Jumlah_negara, SUM(n.nilai_gdp) AS total_gdp,
-> SUM(n.emisi_co2) AS total_emisi_co2
-> FROM Indikator_Negara n
-> JOIN Tahun t ON n.tahun = t.tahun
-> GROUP BY t.dekade
-> ORDER BY t.dekade ASC;

```

dekade	Jumlah_negara	total_gdp	total_emisi_co2
2020s	218	111108606997237	38258727774

1 row in set (0.001 sec)

After Insert

```
MariaDB [Data_Warehouse]> SELECT t.dekade, COUNT(n.id_negara) AS Jumlah_negara, SUM(n.nilai_gdp) AS total_gdp, SUM(n.emisi_co2) AS total_emisi_co2 FROM Indikator_Negara n JOIN Tahun t ON n.tahun = t.tahun GROUP BY t.dekade ORDER BY t.dekade ASC;
```

dekade	Jumlah_negara	total_gdp	total_emisi_co2
2020s	219	136108606997237	43258727774

```
1 row in set (0.002 sec)
```

-- Insert manual untuk tahun 2023 (1 negara doang)

```
INSERT INTO Indikator_Negara VALUES (1, 2023, 'High income', 25000000000000, 85000, 25.5000000, 1, 3300000000, 36, 82.50, 38.50, 1.65, 3, 5000000000, 15.20, 2, 100000000000, 30000, 1, DEFAULT);
```

5. **(Query Optimization)** Mendapatkan data negara dengan gdp per capita dalam range 4500 - 5500

BEFORE

QUERY:

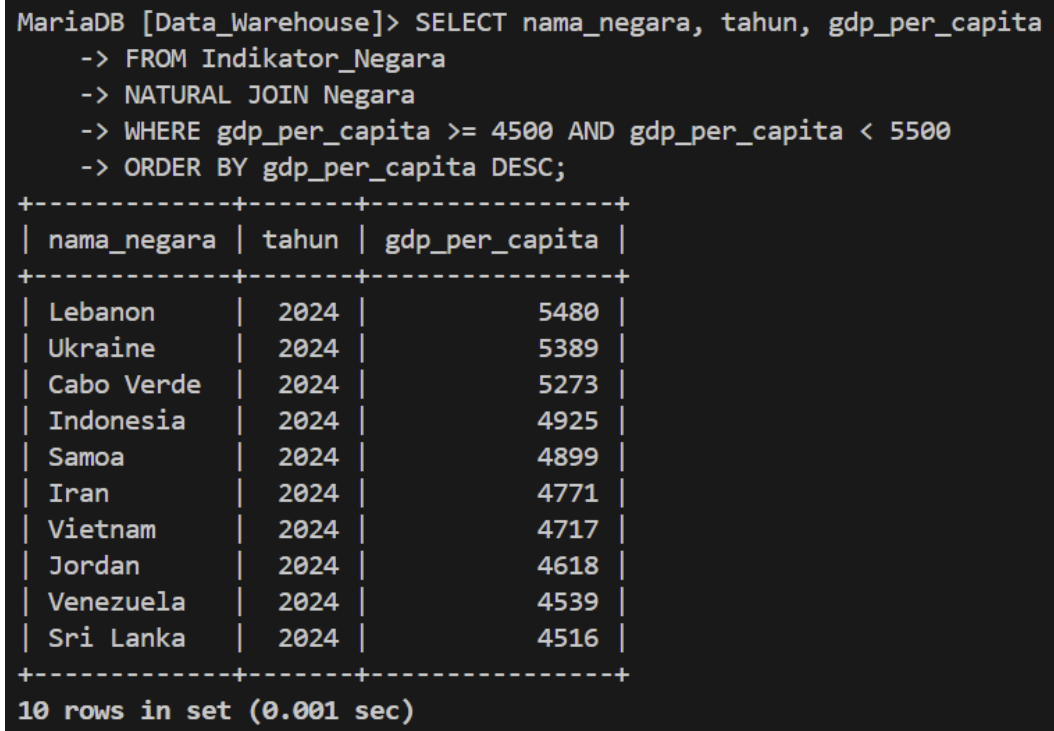
```
SELECT nama_negara, tahun, gdp_per_capita
FROM Indikator_Negara
NATURAL JOIN Negara
WHERE ROUND(gdp_per_capita / 1000) = 5
ORDER BY gdp_per_capita DESC;
```

SCREENSHOT:

```
MariaDB [Data_Warehouse]> SELECT nama_negara, tahun, gdp_per_capita
-> FROM Indikator_Negara
-> NATURAL JOIN Negara
-> WHERE ROUND(gdp_per_capita / 1000) = 5
-> ORDER BY gdp_per_capita DESC;
```

nama_negara	tahun	gdp_per_capita
Lebanon	2024	5480
Ukraine	2024	5389
Cabo Verde	2024	5273
Indonesia	2024	4925
Samoa	2024	4899
Iran	2024	4771
Vietnam	2024	4717
Jordan	2024	4618
Venezuela	2024	4539
Sri Lanka	2024	4516

```
10 rows in set (0.002 sec)
```

AFTER <i>(tidak menggunakan fungsi)</i>
QUERY: SELECT nama_negara, tahun, gdp_per_capita FROM Indikator_Negara NATURAL JOIN Negara WHERE gdp_per_capita >= 4500 AND gdp_per_capita < 5500 ORDER BY gdp_per_capita DESC;
SCREENSHOT:  <pre> MariaDB [Data_Warehouse]> SELECT nama_negara, tahun, gdp_per_capita -> FROM Indikator_Negara -> NATURAL JOIN Negara -> WHERE gdp_per_capita >= 4500 AND gdp_per_capita < 5500 -> ORDER BY gdp_per_capita DESC; +-----+-----+-----+ nama_negara tahun gdp_per_capita +-----+-----+-----+ Lebanon 2024 5480 Ukraine 2024 5389 Cabo Verde 2024 5273 Indonesia 2024 4925 Samoa 2024 4899 Iran 2024 4771 Vietnam 2024 4717 Jordan 2024 4618 Venezuela 2024 4539 Sri Lanka 2024 4516 +-----+-----+-----+ 10 rows in set (0.001 sec) </pre>

6. **(Query Optimization)** Mendapatkan data negara dengan nilai GDP-nya yang terletak di Benua Asia

BEFORE
QUERY: SELECT nama_negara, nilai_gdp, tahun FROM Indikator_Negara i NATURAL JOIN Negara WHERE id_negara in (SELECT id_negara FROM Negara WHERE nama_benua = 'Asia');
SCREENSHOT:

```

MariaDB [Data_Warehouse]> SELECT nama_negara, nilai_gdp, tahun
-> FROM Indikator_Negara i
-> NATURAL JOIN Negara
-> WHERE id_negara in (SELECT id_negara FROM Negara WHERE nama_benua = 'Asia');

```

nama_negara	nilai_gdp	tahun
Afghanistan	17817000000	2024
Armenia	25786585950	2024
Azerbaijan	74315882353	2024
Bahrain	47736702128	2024
Bangladesh	450119424622	2024
Bhutan	3147000000	2024
Brunei	15463134387	2024
Cambodia	46352647035	2024
China	18743803170827	2024
Cyprus	36333022329	2024
Georgia	33776141251	2024
Hong Kong	407106738445	2024
India	3912686168582	2024
Indonesia	1396300098191	2024
Iran	436906331672	2024
Iraq	279641257615	2024
Israel	540379921262	2024
Japan	4026210821147	2024
Jordan	53352289577	2024
Kazakhstan	288406138231	2024
Kuwait	160227273001	2024
Kyrgyzstan	17478259659	2024
Laos	16502933121	2024
Lebanon	29339000000	2024
Macao	50183212761	2024
Malaysia	421972102254	2024
Maldives	6975146349	2024
Mongolia	23586055802	2024
Myanmar	74079772652	2024
Nepal	42914268287	2024
North Korea	16446632290	2024
Oman	106942782835	2024
Pakistan	373071855732	2024
Philippines	461617509782	2024
Qatar	217982967033	2024
Saudi Arabia	1237529866667	2024
Singapore	547386645892	2024
South Korea	1875388000000	2024
Sri Lanka	98963185510	2024
State of Palestine	13711100000	2024
Syria	19993439950	2024
Taiwan	801495000000	2024
Tajikistan	14204575549	2024
Thailand	526411265428	2024
Timor-Leste	1881265333	2024
Turkey	1323254808059	2024
Turkmenistan	64239891739	2024
United Arab Emirates	537078829135	2024
Uzbekistan	114965293467	2024
Vietnam	476388230307	2024
Yemen	8058000000	2024

51 rows in set (0.007 sec)

AFTER (<i>tidak menggunakan subquery</i>)
QUERY: SELECT nama_negara, nilai_gdp, tahun FROM Indikator_Negara i NATURAL JOIN Negara WHERE nama_benua = 'Asia';
SCREENSHOT:

```

MariaDB [Data_Warehouse]> SELECT nama_negara, nilai_gdp, tahun
-> FROM Indikator_Negara i
-> NATURAL JOIN Negara
-> WHERE nama_benua = 'Asia';

```

nama_negara	nilai_gdp	tahun
Afghanistan	17817000000	2024
Armenia	25786585950	2024
Azerbaijan	74315882353	2024
Bahrain	47736702128	2024
Bangladesh	450119424622	2024
Bhutan	3147000000	2024
Brunei	15463134387	2024
Cambodia	46352647035	2024
China	18743803170827	2024
Cyprus	36333022329	2024
Georgia	33776141251	2024
Hong Kong	407106738445	2024
India	3912686168582	2024
Indonesia	1396300098191	2024
Iran	436906331672	2024
Iraq	279641257615	2024
Israel	540379921262	2024
Japan	4026210821147	2024
Jordan	53352289577	2024
Kazakhstan	288406138231	2024
Kuwait	160227273001	2024
Kyrgyzstan	17478259659	2024
Laos	16502933121	2024
Lebanon	29339000000	2024
Macao	50183212761	2024
Malaysia	421972102254	2024
Maldives	6975146349	2024
Mongolia	23586055802	2024
Myanmar	74079772652	2024
Nepal	42914268287	2024
North Korea	16446632290	2024
Oman	106942782835	2024
Pakistan	373071855732	2024
Philippines	461617509782	2024
Qatar	217982967033	2024
Saudi Arabia	1237529866667	2024
Singapore	547386645892	2024
South Korea	1875388000000	2024
Sri Lanka	98963185510	2024
State of Palestine	13711100000	2024
Syria	19993439950	2024
Taiwan	801495000000	2024
Tajikistan	14204575549	2024
Thailand	526411265428	2024
Timor-Leste	1881265333	2024
Turkey	1323254808059	2024
Turkmenistan	64239891739	2024
United Arab Emirates	537078829135	2024
Uzbekistan	114965293467	2024
Vietnam	476388230307	2024
Yemen	8058000000	2024

51 rows in set (0.001 sec)

--

7. **(Query Optimization)** Mendapatkan data negara dengan jumlah komposisi energi > 100 juta, beserta jenisnya

BEFORE
QUERY: SELECT nama_negara, tahun, nama_jenis, jumlah_komposisi FROM Komposisi_Energi NATURAL JOIN Negara NATURAL JOIN Jenis_Energi WHERE jumlah_komposisi / 1000000 > 100;
SCREENSHOT:

```

MariaDB [Data_Warehouse]> SELECT nama_negara, tahun, nama_jenis, jumlah_komposisi
-> FROM Komposisi_Energi
-> NATURAL JOIN Negara
-> NATURAL JOIN Jenis_Energi
-> WHERE jumlah_komposisi / 1000000 > 100;

```

nama_negara	tahun	nama_jenis	jumlah_komposisi
Afghanistan	2017	Oil	67719000000
Albania	2017	Oil	58100000000
Algeria	2017	Oil	836285000000
Angola	2017	Oil	260938000000
Antigua and Barbuda	2017	Oil	10275000000
Argentina	2017	Oil	1536946000000
Armenia	2017	Oil	16603000000
Aruba	2017	Oil	16626000000
Australia	2017	Oil	2387752000000
Austria	2017	Oil	558198000000
Azerbaijan	2017	Oil	197835000000
Bahamas	2017	Oil	42416000000
Bahrain	2017	Oil	131325000000
Bangladesh	2017	Oil	265244000000
Barbados	2017	Oil	22983000000
Belarus	2017	Oil	281370000000
Belgium	2017	Oil	1356044000000
Belize	2017	Oil	8161000000
Benin	2017	Oil	90177000000
Bermuda	2017	Oil	11126000000
Bhutan	2017	Oil	5367000000
Bolivia	2017	Oil	202423000000
Bosnia and Herzegovina	2017	Oil	69914000000
Botswana	2017	Oil	44686000000
Brazil	2017	Oil	6054245000000
British Virgin Islands	2017	Oil	3193000000
Brunei	2017	Oil	31823000000
Bulgaria	2017	Oil	196462000000
Burkina Faso	2017	Oil	45380000000
Burundi	2017	Oil	3087000000
Cabo Verde	2017	Oil	11948000000
Cambodia	2017	Oil	99258000000
Cameroon	2017	Oil	72592000000
Canada	2017	Oil	4944458000000
Cayman Islands	2017	Oil	9514000000
Central African Republic	2017	Oil	5340000000
Chad	2017	Oil	4397000000
Chile	2017	Oil	711780000000
China	2017	Oil	24912052000000
Colombia	2017	Oil	677806000000
Comoros	2017	Oil	3848000000
Congo	2017	Oil	32925000001
Cook Islands	2017	Oil	811000000
Costa Rica	2017	Oil	111736000000
Croatia	2017	Oil	149852000000
Cuba	2017	Oil	352395000000
Cyprus	2017	Oil	113001000000
Czech Republic (Czechia)	2017	Oil	436167000000
Côte d'Ivoire	2017	Oil	88594000000
DR Congo	2017	Oil	42730000000
Denmark	2017	Oil	328061000000
Djibouti	2017	Oil	12074000000
Dominica	2017	Oil	3344000000
Dominican Republic	2017	Oil	244686000000

Papua New Guinea	2017	Renewable & Nuclear	7578484000
Paraguay	2017	Renewable & Nuclear	397226518000
Peru	2017	Renewable & Nuclear	287342497000
Philippines	2017	Renewable & Nuclear	213274315000
Poland	2017	Renewable & Nuclear	55087269000
Portugal	2017	Renewable & Nuclear	198288122000
Puerto Rico	2017	Renewable & Nuclear	2948962000
Qatar	2017	Renewable & Nuclear	817871000
Romania	2017	Renewable & Nuclear	359066406000
Russia	2017	Renewable & Nuclear	3673213530000
Rwanda	2017	Renewable & Nuclear	3063435000
Réunion	2017	Renewable & Nuclear	9771585000
Saint Kitts & Nevis	2017	Renewable & Nuclear	195704000
Saint Lucia	2017	Renewable & Nuclear	500213000
Saint Pierre & Miquelon	2017	Renewable & Nuclear	991000000
Samoa	2017	Renewable & Nuclear	484659000
Sao Tome & Principe	2017	Renewable & Nuclear	215491000
Saudi Arabia	2017	Renewable & Nuclear	1965444000
Senegal	2017	Renewable & Nuclear	5336405000
Serbia	2017	Renewable & Nuclear	107715004000
Sierra Leone	2017	Renewable & Nuclear	810542000
Singapore	2017	Renewable & Nuclear	15665046000
Slovakia	2017	Renewable & Nuclear	246255843000
Slovenia	2017	Renewable & Nuclear	99803901000
Solomon Islands	2017	Renewable & Nuclear	417491000
Somalia	2017	Renewable & Nuclear	628621000
South Africa	2017	Renewable & Nuclear	242275468000
South Korea	2017	Renewable & Nuclear	1577858102000
South Sudan	2017	Renewable & Nuclear	217639000
Spain	2017	Renewable & Nuclear	1390542035000
Sri Lanka	2017	Renewable & Nuclear	42687933000
State of Palestine	2017	Renewable & Nuclear	19485937000
Sudan	2017	Renewable & Nuclear	86446194000
Suriname	2017	Renewable & Nuclear	9713993000
Sweden	2017	Renewable & Nuclear	1459423150000
Switzerland	2017	Renewable & Nuclear	577310930000
Syria	2017	Renewable & Nuclear	5485856000
Taiwan	2017	Renewable & Nuclear	332922669000
Tajikistan	2017	Renewable & Nuclear	152056535000
Tanzania	2017	Renewable & Nuclear	21779707000
Thailand	2017	Renewable & Nuclear	361425308000
Togo	2017	Renewable & Nuclear	6603410000
Tunisia	2017	Renewable & Nuclear	5957307000
Turkey	2017	Renewable & Nuclear	816328927000
Turks and Caicos	2017	Renewable & Nuclear	303000000
Uganda	2017	Renewable & Nuclear	33092787000
Ukraine	2017	Renewable & Nuclear	997073230000
United Arab Emirates	2017	Renewable & Nuclear	6997779000
United Kingdom	2017	Renewable & Nuclear	1605071777000
United States	2017	Renewable & Nuclear	18237625460000
Uruguay	2017	Renewable & Nuclear	117634319000
Uzbekistan	2017	Renewable & Nuclear	74490562000
Vanuatu	2017	Renewable & Nuclear	571838000
Venezuela	2017	Renewable & Nuclear	588615837000
Vietnam	2017	Renewable & Nuclear	654067685000
Yemen	2017	Renewable & Nuclear	1142049000
Zambia	2017	Renewable & Nuclear	110475040000
Zimbabwe	2017	Renewable & Nuclear	42209625000

625 rows in set (0.003 sec)

AFTER (use indexing)

QUERY:

```
CREATE INDEX idx_komposisi ON Komposisi_Energi (jumlah_komposisi);
```

```
SELECT n.nama_negara, k.tahun, j.nama_jenis, k.jumlah_komposisi
FROM Komposisi_Energi k
JOIN Negara n ON k.id_negara = n.id_negara
JOIN Jenis_Energi j ON k.id_jenis = j.id_jenis
WHERE k.jumlah_komposisi > 100000000;
```

SCREENSHOT:

```
MariaDB [Data_Warehouse]> CREATE INDEX idx_komposisi ON Komposisi_Energi (jumlah_komposisi);
Query OK, 0 rows affected (0.123 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

```
MariaDB [Data_Warehouse]> SELECT n.nama_negara, k.tahun, j.nama_jenis, k.jumlah_komposisi FROM Komposisi_Energi k
-> JOIN Negara n ON k.id_negara = n.id_negara
-> JOIN Jenis_Energi j ON k.id_jenis = j.id_jenis
-> WHERE k.jumlah_komposisi > 100000000;
```

nama_negara	tahun	nama_jenis	jumlah_komposisi
Afghanistan	2017	Oil	67719000000
Albania	2017	Oil	58100000000
Algeria	2017	Oil	836285000000
Angola	2017	Oil	260938000000
Antigua and Barbuda	2017	Oil	10275000000
Argentina	2017	Oil	1536946000000
Armenia	2017	Oil	16603000000
Aruba	2017	Oil	16626000000
Australia	2017	Oil	2387752000000
Austria	2017	Oil	558198000000
Azerbaijan	2017	Oil	197835000000
Bahamas	2017	Oil	42416000000
Bahrain	2017	Oil	131325000000
Bangladesh	2017	Oil	265244000000
Barbados	2017	Oil	22983000000
Belarus	2017	Oil	281370000000
Belgium	2017	Oil	1356044000000
Belize	2017	Oil	8161000000
Benin	2017	Oil	90177000000
Bermuda	2017	Oil	11126000000
Bhutan	2017	Oil	5367000000
Bolivia	2017	Oil	202423000000
Bosnia and Herzegovina	2017	Oil	69914000000
Botswana	2017	Oil	44686000000
Brazil	2017	Oil	6054245000000
British Virgin Islands	2017	Oil	3193000000
Brunei	2017	Oil	31823000000
Bulgaria	2017	Oil	196462000000
Burkina Faso	2017	Oil	45380000000
Burundi	2017	Oil	3087000000
Cabo Verde	2017	Oil	11948000000
Cambodia	2017	Oil	99258000000
Cameroon	2017	Oil	72592000000
Canada	2017	Oil	4944458000000
Cayman Islands	2017	Oil	9514000000
Central African Republic	2017	Oil	5340000000
Chad	2017	Oil	4397000000
Chile	2017	Oil	71178000000
China	2017	Oil	24912052000000
Colombia	2017	Oil	677806000000
Comoros	2017	Oil	3848000000
Congo	2017	Oil	32925000001
Cook Islands	2017	Oil	811000000
Costa Rica	2017	Oil	111736000000
Croatia	2017	Oil	149852000000
Cuba	2017	Oil	352395000000

Russia	2017	Renewable & Nuclear	3673213530000
Rwanda	2017	Renewable & Nuclear	3063435000
Réunion	2017	Renewable & Nuclear	9771585000
Saint Kitts & Nevis	2017	Renewable & Nuclear	195704000
Saint Lucia	2017	Renewable & Nuclear	500213000
Saint Pierre & Miquelon	2017	Renewable & Nuclear	991000000
Samoa	2017	Renewable & Nuclear	484659000
Sao Tome & Principe	2017	Renewable & Nuclear	215491000
Saudi Arabia	2017	Renewable & Nuclear	1965444000
Senegal	2017	Renewable & Nuclear	5336405000
Serbia	2017	Renewable & Nuclear	107715004000
Sierra Leone	2017	Renewable & Nuclear	810542000
Singapore	2017	Renewable & Nuclear	15665046000
Slovakia	2017	Renewable & Nuclear	246255843000
Slovenia	2017	Renewable & Nuclear	99803901000
Solomon Islands	2017	Renewable & Nuclear	417491000
Somalia	2017	Renewable & Nuclear	628621000
South Africa	2017	Renewable & Nuclear	242275468000
South Korea	2017	Renewable & Nuclear	1577858102000
South Sudan	2017	Renewable & Nuclear	217639000
Spain	2017	Renewable & Nuclear	1390542035000
Sri Lanka	2017	Renewable & Nuclear	42687933000
State of Palestine	2017	Renewable & Nuclear	19485937000
Sudan	2017	Renewable & Nuclear	86446194000
Suriname	2017	Renewable & Nuclear	9713993000
Sweden	2017	Renewable & Nuclear	1459423150000
Switzerland	2017	Renewable & Nuclear	577310930000
Syria	2017	Renewable & Nuclear	5485856000
Taiwan	2017	Renewable & Nuclear	332922669000
Tajikistan	2017	Renewable & Nuclear	152056535000
Tanzania	2017	Renewable & Nuclear	21779707000
Thailand	2017	Renewable & Nuclear	361425308000
Togo	2017	Renewable & Nuclear	6603410000
Tunisia	2017	Renewable & Nuclear	5957307000
Turkey	2017	Renewable & Nuclear	816328927000
Turks and Caicos	2017	Renewable & Nuclear	303000000
Uganda	2017	Renewable & Nuclear	33092787000
Ukraine	2017	Renewable & Nuclear	997073230000
United Arab Emirates	2017	Renewable & Nuclear	6997779000
United Kingdom	2017	Renewable & Nuclear	1605071777000
United States	2017	Renewable & Nuclear	18237625460000
Uruguay	2017	Renewable & Nuclear	117634319000
Uzbekistan	2017	Renewable & Nuclear	74490562000
Vanuatu	2017	Renewable & Nuclear	571838000
Venezuela	2017	Renewable & Nuclear	588615837000
Vietnam	2017	Renewable & Nuclear	654067685000
Yemen	2017	Renewable & Nuclear	1142049000
Zambia	2017	Renewable & Nuclear	110475040000
Zimbabwe	2017	Renewable & Nuclear	42209625000

625 rows in set (0.002 sec)

8. Mengecek *scheduling*

UPDATING (just for showcase)

QUERY:
UPDATE Data_Storing.Indikator_GDP
SET nilai_gdp = nilai_gdp + 100
WHERE id_negara = 1 AND tahun = 2024;

SCREENSHOT:

```
MariaDB [Data_Warehouse]> UPDATE Data_Storing.Indikator_GDP
-> SET nilai_gdp = nilai_gdp + 100
-> WHERE id_negara = 1 AND tahun = 2024;
Query OK, 1 row affected (0.028 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

AFTER UPDATE

QUERY:
SELECT * FROM Indikator_Negara LIMIT 5;

SCREENSHOT:

```
MariaDB [Data_Warehouse]> SELECT * FROM Indikator_Negara LIMIT 5;
```

id_negara	tahun	jenis_kategori_ekonomi	nilai_gdp	gdp_per_capita	persentase_gdp_dunia	rank_gdp	jumlah_populasi	kepadatan_penduduk	persentase_penduduk_urban	usia_medan	tingkat_fertilitas	rank_populasi	emisi_co2	emisi_co2_per_capita	rank_co2	konsumsi_energi_total	konsumsi_energi_per_capita	rank_energi	created_at	updated_at	
1	2024	Low Income	17817000100	418	0.0200000	136	42647492	65	26.50	17.10	4.76	36	12068415	0.29	104	0.0200000	115	2791765	102	2026-07-22 14:18:12	2026-07-22 14:23:12
2	2024	Upper middle income	27177735528	10012	0.0200000	115	2791765	102	69.00	36.70	1.34	144	4243706	1.44	142	0.2400000	52	46814308	20	2026-07-22 14:18:12	2026-07-22 14:18:12
3	2024	Upper middle income	263619794507	5631	0.2400000	52	46814308	20	74.00	28.40	2.72	32	186358382	4.07	31	0.0008580	205	46765	234	2026-07-22 14:18:12	2026-07-22 14:18:12
4	2024	High income	871000000	18017	0.0008580	205	46765	234	83.20	43.50	2.27	210	NULL	NULL	NULL	0.0036000	176	81938	174	2026-07-22 14:18:12	2026-07-22 14:18:12
5	2024	High income	4039844390	49304	0.0036000	176	81938	174	83.20	43.50	1.09	201	NULL	NULL	NULL	NULL	NULL	NULL	NULL	2026-07-22 14:18:12	2026-07-22 14:18:12

5 rows in set (0.001 sec)

-- waktu updated_at untuk id_negara = 1 ikut berubah